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Modular personal storage system and related modular components

Abstract

A modular personal storage system can include a personal storage container that can include a first outer wall including a paramagnetic metal or a diamagnetic metal, a second outer wall including the paramagnetic metal or the diamagnetic metal, where the second outer wall is positioned opposite to and spaced apart from the first outer wall to define a container volume between the first outer wall and the second outer wall. At least two permanent magnetic elements can be spaced apart from one another and at least partially located within the container volume.

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Background/Summary

PRIORITY [0001] This application is a continuation under 35 U.S.C. § 120 of U.S. patent application Ser. No. 18/057,406, filed 21 Nov. 2022, which claims the benefit under 35 U.S.C. § 119(c) of the following provisional applications: U.S. Provisional Patent Application No. 63/281,719, filed 21 Nov. 2021, U.S. Provisional Patent Application 63/282,089, filed 22 Nov. 2021, and U.S. Provisional Patent Application 63/282,321 filed 23 Nov. 2021, which are incorporated herein by reference.

BACKGROUND

[0002] It is known to provide a wallet that may be removably attachable to a mobile electronic device or a case for the mobile electronic device. Such wallets may be used to hold one or more objects (e.g., credit cards, identification cards, business cards, pictures, cash, keys, or the like) associated with (e.g., owned by) a user of the accessory (in many cases, the user will be using the mobile electronic device as well).

[0003] When the wallet is attached to the mobile electronic device or the case therefor, the user can securely and easily store and carry such objects along with his/her mobile electronic device. At the same time, the wallet can be removed from the mobile electronic device (or the case) when desired (e.g., to reduce the profile of the mobile electronic device, in order to enable wireless charging).

SUMMARY OF PARTICULAR EMBODIMENTS

[0004] Embodiments according to the inventive concept can provide modular personal storage systems and related modular components. Pursuant to these embodiments, a modular personal storage system can comprise a personal storage container that can include a first outer wall including a paramagnetic metal or a diamagnetic metal, a second outer wall including the paramagnetic metal or the diamagnetic metal, where the second outer wall is positioned opposite to and spaced apart from the first outer wall to define a container volume between the first outer wall and the second outer wall. At least two permanent magnetic elements can be spaced apart from one another and at least partially located within the container volume.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0005] FIGS. 1A-C are schematic illustrations of a modular personal storage system including personal storage containers magnetically removably coupled to one another and removable covers to provide a customized configuration of for both function and appearance of the modular personal storage system in some embodiments according to the inventive concept.

[0006] FIGS. 2A-J are schematic illustrations of an exemplary personal storage container that can be included in the modular personal storage system of FIGS. 1A-C configured to carry a plurality of payment cards, or objects having an equivalent form-factor, for secure storage and retrieval in some embodiments according to the inventive concept.

[0007] FIGS. 3A-D are schematic illustrations of an exemplary personal storage container that can be included in the modular personal storage system of FIGS. 1A-C configured to carry a utility tool or equivalent object for secure storage and retrieval in some embodiments according to the inventive concept.

[0008] FIGS. 4A-D are schematic illustrations of an exemplary cover that can be included in the

personal storage system of FIGS. 1A-C and configured to carry loose items, such as currency, under an elastic member that extends around a central portion of the cover for secure storage and retrieval in some embodiments according to the inventive concept.

[0009] FIGS. 5A-D are schematic illustrations of an exemplary cover that can be included in the personal storage system of FIGS. 1A-C to customize the appearance of the outer surfaces of the modular personal storage system in some embodiments according to the inventive concept.

[0010] FIG. 6 is a schematic illustration of an exemplary cover that can be included in the personal storage system of FIGS. 1A-C and configured to carry an identification, or objects having an equivalent form-factor, on an outside of the personal storage system for secure storage and retrieval in some embodiments according to the inventive concept.

[0011] FIGS. 7A-B are schematically illustrations of a magnetic phone mount configured to securely magnetically couple the personal storage system of FIGS. 1A-C to a mobile device in some embodiments according to the inventive concept.

[0012] FIG. 7C is a schematic illustration of the magnetic phone mount of FIGS. 7A-B coupled to a mobile device and configured to magnetically couple to the modular personal storage system including the personal storage containers and covers disclosed herein in some embodiments according to the inventive concept.

[0013] FIG. 7D is a schematic illustration of a personal storage container and cover magnetically coupled to the mobile device via the magnetic phone mount of FIGS. 7A-B in some embodiments according to the inventive concept.

DESCRIPTION OF EXAMPLE EMBODIMENTS

[0014] As appreciated by the present inventors, a modular approach to personal storage can enable users to select from a range of personal storage containers that best meet their needs both from a functional and decorative standpoint. For example, users can select a single personal storage container to carry a number of payment cards, but can add one or more personal storage containers when additional cards may be needed. Alternatively, users may select a personal storage container that is configured to carry items other than payment cards, such as a tool or other objects.

Furthermore, the magnetic coupling of the personal storage containers can make adding personal storage containers more convenient by bringing the new personal storage container into contact with the existing one so that stack of the two (or more) personal storage containers can function as a larger single personal storage system.

[0015] In some embodiments according to the present inventive concept, the magnetic coupling of the personal storage containers can allow the user to re-arrange the order or orientation of existing containers to be in a more desirable configuration. Still further, in some embodiments according to the present inventive concept, the personal storage containers be coupled together to a mobile electronic device so as to be more conveniently carried but still provide easy access to the items carried in the personal storage containers. In some embodiments according to the present inventive concept, the user can select from among a variety of personal storage covers to conveniently customize the appearance of the personal storage containers. In particular, the covers can include ferromagnetic patterns that can magnetically couple to the personal storage containers that are to be customized. When the user desires to change the outward appearance of the carried personal storage containers, the new cover can be magnetically coupled to the outer surface (or surfaces) of the outermost personal storage containers included in the stack. In still further embodiments according to the inventive concept, the covers can include functional features. For example, in some embodiments according to the inventive concept, a cover can include an elastic element that can allow the user to carry otherwise loose items by inserting the items (such as currency) between the cover and the clastic element so that the items are compressively held to the cover. In still further embodiments according to the present inventive concept, the cover can include a tray that is recessed into the surface of the cover with a lip that overhangs the tray at the edges. The tray can be configured to carry payment cards, identification and/or other items having a form-factor that fits

within the tray by compressing the items between the bottom of the tray and the lip.

[0016] FIGS. 1A-C are schematic illustrations of a modular personal storage system **100** including personal storage containers **105** magnetically removably coupled to one another and removable covers **110** to provide a customizable stack that can be configured for both function and appearance of the modular personal storage system **100** in some embodiments according to the inventive concept.

[0017] Referring to FIG. 1A, the modular personal storage system **100** can include one or more personal storage containers **105** that can be magnetically coupled together as described herein. A first personal storage container **105A** can function as a wallet that is configured to carry a plurality of payment cards or items with a similar form factor. As further described herein, the personal storage container **105A** can include a lever that is configured to push the items stored inside the personal storage container **105A** through an opening so that they may be accessed by the user. As further shown in FIG. 1A, a second personal storage container **105B** can be removably magnetically coupled to the personal storage container **105A** and is configured to carry a tool. Although the modular personal storage system **100** includes two personal storage containers **105**, it will be understood that fewer or more personal storage containers may be used in a stack.

[0018] As further shown in FIG. 1A, the modular personal storage system **100** can include removable decorative covers **110** that can be magnetically coupled to either outer surface of the personal storage containers **105** included in the stack. As shown, a removable decorative covers **110A** can be selected by the user to customize the outward appearance of the stack. As further shown in FIG. 1A, the stack can also include a removable decorative/cash band cover **110B** which can be magnetically coupled to the lower outer surface of the stack. As described herein, the cash band can be an elastic member that is configured to hold otherwise loose objects, such as currency, securely to the outer surface of the stack that makes up the modular personal storage system **100**.

[0019] It will be further understood that each of the items included in the modular personal storage system **100** can be magnetically coupled to the directly adjacent component in the stack via permanent magnets included in the personal storage containers **105** or corresponding ferromagnetic materials patterned into the interior portions of the removable covers **110**. It will be further understood that although FIG. 1A shows two types of decorative covers other types of decorative covers can be selected for coupling to the stack as described herein.

[0020] FIG. 1B is a perspective view of the modular personal storage system **100** shown in FIG. 1A having the decorative removable cover **110A** coupled to a top surface of the second personal storage container **105B** (e.g., the top container) in some embodiments according to the inventive concept. It will be understood that the removable decorative cover **110** can be removed and replaced with an alternative decorative removable cover **110A** to change the outward appearance of the stack of personal storage containers **105**.

[0021] FIG. 1C is a schematic side view representation of the personal storage containers **105** and the removable decorative covers **110** shown in FIG. 1A positioned relative to one another to form a stack **111**. As further shown in Figure 1C, the removable decorative cover **110A** can be magnetically coupled to the first personal storage container **105A** whereas the removable decorative cover **110B** can be coupled to the second personal storage container **105B**. It will be understood that in some embodiments according to the inventive concept, the order of the personal storage containers **105** shown in the stack **111** can be changed. Moreover, the removable decorative covers **110** can be magnetically coupled to either of the personal storage containers **105**. As further shown in Figure 1C, when the personal storage containers **105** and the decorative removable covers **110** are brought into contact with one another in the relative positions shown, each of the components can be magnetically coupled to the contacting component as shown.

[0022] FIGS. 2A-H are schematic illustrations of an exemplary personal storage container **105A** included in the modular personal storage system **100** of FIG. 1A configured to carry a plurality of payment cards, or objects having a similar form-factor, for secure storage and retrieval in some

embodiments according to the inventive concept. As shown in FIG. 2A, the personal storage container **105A** includes a first outer wall **210** surrounded by an outer side wall **226** which extends around a perimeter of the personal storage container **105A** and leaves an opening **239** therein that is configured to allow the insertion and removal of a plurality of items stored within (such as a plurality of payment cards).

[0023] As further shown in FIG. 2A, the first outer wall **210** includes first and second separator bumpers **220** positioned adjacent to where the first outer wall **210** meets the outer side wall **226** at opposing ends of the first outer wall **210**. It will be understood that in some embodiments according to the inventive concept, the separator bumpers **220** are configured to maintain a small separation between adjacent personal storage containers **105** when magnetically coupled together so that the magnetic coupling therebetween may be more easily overcome to separate the personal storage containers **105** from one another. As shown in FIG. 2J, the separator bumpers **220** can have opposing straight portions of length W and curved end portions, with radii $R2$, that join the opposing straight portions. In some embodiments according to the inventive concept, the length W can be about 7.9 mm and the radii $R2$ can be about 0.05 mm. In some embodiments according to the inventive concept, the separator bumpers **220** can protrude about 0.15 mm from the surface of the outer wall.

[0024] Still referring to FIG. 2A, a lever **230** is located at a corner of the personal storage container **105A** on the outer side wall **226**. As described herein, the lever **230** is configured to push the plurality of items stored inside the personal storage container **105** out through the opening **239** when the lever is activated by the user. In some embodiments according to the present inventive concept, the first outer wall **210** includes an alignment opening **225** that exposes the interior of the personal storage container **105**. The alignment opening **225** can be used to visually and/or mechanically align the personal storage container **105** with another personal storage container or removeable decorative cover **110**, which may include a correspondingly shaped alignment protrusion on the surface which is configured for insertion into the alignment opening **225** to align the two components in two dimensions magnetically couple with the personal storage container **105**. In some embodiments according to the inventive concept, the alignment opening **225** is stadium shaped to include two spaced-apart straight portions having a length L and two curved portions at opposing ends of the two spaced-apart straight portions with radii of RI configured to join the two spaced-apart straight portions. In some embodiments according to the inventive concept, alignment opening **225** can include opposing straight portions having lengths of about 16.5 mm and curved portions having a respective radius of about 0.9 mm on each end to connect with the alignment opening to form the stadium shape. In some embodiments according to the inventive concept, alignment opening **225** can be sized about 0.3 mm larger than the dimensions of the alignment protrusion.

[0025] Referring to FIGS. 2B and 2C, the personal storage container **105A** includes a second outer wall **215** surrounded by the outer side wall **226**. The second outer wall is spaced apart from, and opposite to, the first outer wall **210**. The space bounded by the perimeter of the outer side wall **226** and the space between the first and second outer walls **210** and **215** define a container volume **213** that includes many of the components of the personal storage containers **105** described herein.

[0026] As illustrated in the cross-sectional view of FIG. 2C taken along line C-C' in FIG. 2B, the interior sidewall **218** follows the perimeter of the container volume **213** that provides the portion of the container volume that can be occupied by the contents stored in the personal storage container **105** and the structures that provide for the operation and configuration of the personal storage container **105**. It will be understood that the container volume **213** is defined to also include the components of the personal storage container **105** that are outside the interior sidewall **218**.

[0027] For example, as shown in FIG. 2C, the container volume **213** also includes space for a plurality of permanent magnetic elements **250** that are spaced apart along the perimeter for three contiguous outer side walls **226** in the container volume. It will be understood that the permanent

magnetic elements **250** can be seated in recesses formed inside the container volume **213** to remain stationary when subjected to forces resulting from magnetic coupling of the personal storage container **105** to other personal storage containers **105** and/or removeable decorative covers **110**. In some embodiments according to the inventive concept, the personal storage container **105** can include six permanent magnetic elements **250** spaced apart along the perimeter.

[0028] In some embodiments according to the inventive concept, the first and second outer walls **210** and **215** can include a diamagnetic or paramagnetic metal or an alloy that includes a diamagnetic or paramagnetic metal. As appreciated by the present inventors, a metal has mobile electrons which can respond to magnetic fields. Accordingly, in the presence of a constant magnetic field (generated by a permanent magnetic element), the mobile electrons (or conduction electrons) in their random motions will orbit about the applied magnetic field. That orbital motion can create a magnetic field, inside the metal, that is oriented in the same direction as the applied magnetic field (paramagnetic) or in the opposite direction as the applied magnetic field (diamagnetic).

[0029] Whether a metal is slightly diamagnetic or slightly paramagnetic depends on which of those two effects is dominant in the particular metal that is exposed to the magnetic field. For example, aluminum is slightly paramagnetic, so inside the aluminum, the magnetic field applied to the aluminum is slightly increased. But, in general, the magnetic field penetrates the aluminum, so the aluminum does not block the magnetic field. Some examples of other paramagnetic metals include tungsten and magnesium. Some examples of diamagnetic metals include copper, zinc, silver, and gold.

[0030] Accordingly, as appreciated by the present inventors, the magnetic field generated by the permanent magnetic elements **250** inside the container volume **213** can pass through the first and second outer walls **210** **215** so that another personal storage container **105** (or a removeable decorative cover **110**) can magnetically couple to the personal storage container **105**. Moreover, because the poles of the permanent magnetic elements **250** of the personal storage containers **105** are facing in opposite directions, additional personal storage containers **105** can be added to the modular personal storage system **100** by placing the outer walls of those containers with opposing poles facing one another. For example, referring to Figure IC, if the south poles of the permanent magnetic elements **250** in the personal storage container **105A** face downward and the north poles face upward, the second personal storage container **105B** will magnetically couple to the personal storage container **105A** by the magnetic coupling between the opposing poles of the respective magnetic elements **250** in the first and second personal storage containers **105A** and **105B**. Additional personal storage containers may be added to the stack by adhering to the same magnetic pole orientation for each additional personal storage container **105**.

[0031] FIG. 2D is a schematic illustration of the personal storage container **105A** with the first outer wall **210** removed so that the interior of the personal storage container **105A** within the container volume **213** is visible in some embodiments according to the inventive concept. According to FIG. 2D, the inside surface of the second outer wall **215** includes ridges **257** that are raised relative to the interior surface thereof. The ridges **257** can reduce the friction between the lowermost payment card stored in the container volume **213** so that the card does not stick inside the container volume **213** when being removed.

[0032] As further shown in FIG. 2D, the lever **230** is located at a corner of the personal storage container **105A** and is connected to a hinge **235** just inside the container volume **213**. The hinge **235** is connected to a finger **240** that has a staircase shaped edge **245** that is configured to contact the edges of the stored payment cards when the lever **230** is moved about the hinge **235** to rotate the finger **240** toward the opening **239** so the staircase shaped edge **245** pushes the cards toward the opening **239** by different amounts so that when the cards are pushed out of the opening **239**, the cards are fanned out so as to be individually accessible by the user.

[0033] As further shown in FIG. 2D first and second deformable bumpers **260** and **265** protrude from the interior sidewall **218** on both sides of the opening **239** and are configured to contact edges

of the plurality of payment cards **270** stored within the storage container **105A** so as to deform relative to when the first and second deformable bumpers **260** and **265** are not in contact with cards **270** stored in the container volume **213**. As further shown in FIG. 2D, the first deformable bumpers **260** are located closest to the opening **239** relative to the second deformable bumpers **265**. In some embodiments according to the inventive concept the first deformable bumpers **260** can protrude from the interior sidewall **218** by an amount P , which can be about 0.5 mm as shown in FIG. 21. In some embodiments according to the inventive concept an outer surface of the second deformable bumpers **265** and **260** can have a radius of $R3$, which in some embodiments can be about 0.073 mm. In some embodiments according to the inventive concept the second deformable bumpers **265** can protrude from the interior sidewall **218** by a different amount or by the same amount P to securely hold the payment cards in place inside the container volume **213** as shown in FIGS. 2E and 2F, for example. In some embodiments according to the inventive concept, the first and second deformable bumpers **260** **265** can be separated by a distance D . In some embodiments according to the inventive concept, the distance D can be about 88 mm. In some embodiments according to the inventive concept, the first and second deformable bumpers **260** **265** can be formed of silicone. Other deformable materials may also be used.

[0034] FIG. 2E is a schematic illustration of the interior of the personal storage container **105A** with a plurality of payment cards **270** stored therein in some embodiments according to the inventive concept. FIG. 2F is a cross-sectional view of the personal storage container **105A**. When the lever **230** is rotated about the hinge **235**, the staircase shaped edge **245** pushes against the edges of the plurality of payment cards to push the cards **270** through the opening **239** so that the edges of the cards **270** are each exposed for convenient access by the user.

[0035] FIG. 2G is a schematic illustration of the personal storage container **105A** with the first outer wall **210** in place and in the same configuration as that shown in FIG. 2F. FIG. 2H is a schematic side view of the personal storage container **105A** showing the first outer wall **210** opposite the 2nd outer wall **215** being coupled together by the side wall **226** that extends around the perimeter to define the container volume **213** as shown in FIGS. 2A-G in some embodiments according to the inventive concept.

[0036] FIGS. 3A-D are schematic illustrations of an exemplary personal storage container **105B** that can be included in the modular personal storage system **100** of FIGS. 1A-C configured to carry a utility tool **301** or equivalent object for secure storage and retrieval in some embodiments according to the inventive concept. According to FIG. 3A, the second personal storage container **105B** includes a first outer wall **310** and a spaced apart second outer wall **315** that defines the container volume **213** which is configured to securely store the utility tool **301**. The utility tool **301** can be inserted or removed from the container volume **213** via an opening **379**. As further shown in FIG. 3A, an alignment opening **225** is also included in the first and second outer walls **310** and **315**, which can have the stadium shape as described herein.

[0037] Referring to FIG. 3B, the utility tool **301** is withdrawn from the container volume **213** through the opening **379**. Referring to FIG. 3C, the second personal storage container **105B** is shown securely storing the utility tool **301** and being magnetically coupled to the first personal storage container **105A**.

[0038] FIG. 3D is a schematic illustration of the second personal storage container **105B** magnetically coupled to the first storage container **105A** and having the first outer wall **310** removed to expose the container volume **213** used to store the utility tool **301** in some embodiments according to the present inventive concept. As further shown in FIG. 3D, the container volume **213** includes eight permanent magnetic elements **350** with two elements positioned on both sides of the corners of the second personal storage container **105B** having the south poles facing downward and located directly opposite the upward facing north poles of the six permanent magnetic elements **250** inside the first personal storage container **105A**.

[0039] It will be understood that because the first personal storage container **105A** includes an

opening 379, the first personal storage container 105A includes fewer permanent magnetic elements 250 compared to those in the second personal storage container 105B. Accordingly, it will be understood that the second personal storage container 105B can be magnetically coupled to either side of the first personal storage container 105A due to the additional permanent magnetic elements therein.

[0040] As further shown in FIG. 3D, the utility tool 301 can be securely stored within the container volume 213 in a tool recess 355 and can be retained by opposing latches 390 in some embodiments according to the inventive concept. In operation, the opposing latches 390 press against the handle of the tool 301 at a point where the diameter is less than the maximum diameter with sufficient compressive force such that the utility tool 301 remains inside the tool recess 355 in all orientations. When the user withdraws the utility tool 301 from the tool recess 355, the opposing latches 390 are pressed outward due to the increased diameter portion of the handle being drawn past the opposing latches 390 to allow the utility tool 301 to be withdrawn from the tool recess 355.

[0041] FIGS. 4A-D are schematic illustrations of an exemplary cover 110B that can be included in the modular personal storage system 100 of FIGS. 1A-C and configured to carry loose items, such as currency, under an elastic member 411 that extends around a central portion of the cover 110B having a notch 407 therein for secure storage and retrieval of the loose items in some embodiments according to the inventive concept. FIG. 4A illustrates an outside surface 419 that can include decorative features or a decorative finish to customize the appearance of the stack. FIG. 4B is a schematic illustration of the inner surface 412 of the exemplary cover 110B including and alignment protrusion 423 that is configured for insertion into the alignment opening 225 in a personal storage container 105 to which the exemplary cover 110B is to be magnetically coupled.

[0042] FIG. 4C illustrates a perspective view of the exemplary cover 110B having the alignment protrusion 423 aligned to and recessed into the alignment opening 225 of the first personal storage container 105A to magnetically couple to exemplary cover 110B in some embodiments.

[0043] FIG. 4D is a schematic illustration of a ferromagnetic pattern 427 located inside the exemplary cover 110B between the inner surface 412 and the outside surface 419 in some embodiments. As further shown in FIG. 4D, the ferromagnetic pattern 427 is located to face the outer wall of the personal storage container 105 to which the exemplary cover 110B is to be magnetically coupled. In some embodiments, the ferromagnetic pattern 427 is shaped to be directly opposite the corresponding permanent magnetic elements 250 located adjacent to the corners of the personal storage container 105. In some embodiments, the ferromagnetic pattern 427 can be a continuous sheet of ferromagnetic material. Other patterns may also be used. In some embodiments, the ferromagnetic pattern 427 can be steel. It will be further understood that the ferromagnetic pattern 427 can be used in other covers 110 and identification carriers that are configured to magnetically couple to a personal storage container 105 but may not include permanent magnetic elements.

[0044] FIGS. 5A-D are schematic illustrations of an exemplary cover 110A that can be included in the modular personal storage system 100 of FIGS. 1A-C in some embodiments according to the inventive concept. FIG. 5A illustrates an outside surface 519 that can include decorative features or a decorative finish to customize the appearance of the stack. FIG. 5B is a schematic illustration of the inner surface 512 of the exemplary cover 110B including and alignment protrusion 523 that is configured for insertion into the alignment opening 225 in a personal storage container 105 to which the exemplary cover 110A is to be magnetically coupled.

[0045] FIG. 5C illustrates a perspective view of the exemplary cover 110A having the alignment protrusion 523 aligned to and recessed into the alignment opening 225 of the first personal storage container 105A to magnetically couple to exemplary cover 110A in some embodiments.

[0046] FIG. 5D is a schematic illustration of a ferromagnetic pattern 427 located inside the exemplary cover 110A between the inner surface 512 and the outside surface 519 in some embodiments. As further shown in FIG. 5D, the ferromagnetic pattern 427 is located to face the

outer wall of the personal storage container **105** to which the exemplary cover **110A** is to be magnetically coupled. In some embodiments, the ferromagnetic pattern **527** is shaped to be directly opposite the corresponding permanent magnetic elements **250** located adjacent to the corners of the personal storage container **105**. In some embodiments, the ferromagnetic pattern **427** can be a continuous sheet of ferromagnetic material. Other patterns may also be used. In some embodiments, the ferromagnetic pattern **427** can be steel. It will be further understood that the ferromagnetic pattern **427** can be used in other covers **110** and identification carriers that are configured to magnetically couple to a personal storage container **105** but may not include permanent magnetic elements.

[0047] FIG. **6** is a schematic illustration of an exemplary cover **110** that can be included in the modular personal storage system **100** of FIGS. **1A-C** and configured to carry an identification, or objects having an equivalent form-factor, on an outside of the modular personal storage system **100** for secure storage and retrieval in some embodiments according to the inventive concept.

According to FIG. **6**, cover **110** can magnetically couple to a personal storage container **105**. The cover **110** can also include a tray **680** that is recessed below a tray edge **622** of the surface of the cover **110** and is surrounded by a lip **662** that extends from the tray edge **622** over a bottom surface **699** of the tray **680**. A card, such as an ID, can be pushed through an opening **618** of the tray **680** so that the edge of the card is captured beneath the lip **662** so that the card can be held securely in the tray **680**. In some embodiments, the bottom surface **699** can include spaced apart raised ridges **690** that extend along the length of the bottom surface **699**. The raised ridges **690** can reduce friction between the card and the bottom surface **699** so that the card may be more easily inserted and removed from the tray **680**.

[0048] It will be understood that a ferromagnetic pattern **427** can be located inside the exemplary cover **110** similar to that described above in reference to FIGS. **5A-D** in some embodiments.

[0049] FIGS. **7A-B** are schematically illustrations of a magnetic phone mount **700** configured to magnetically couple the modular personal storage system **100** of FIGS. **1A-C** securely to a mobile device in some embodiments according to the inventive concept. According to FIG. **7A**, the phone mount includes an outer surface **715** and a wireless charging opening **715** to allow access to the wireless charging interface of a mobile device to which the magnetic phone mount **700** is coupled in some embodiments. It will be understood that the magnetic phone mount **700** can be, for example, compatible with an iPhone MAGSAFE wireless charging standard available from Apple, Inc. of Cupertino, CA.

[0050] FIG. **7B** is a schematic illustration of an interior of the magnetic phone mount **700** in some embodiments. According to FIG. **7B**, the interior of the magnetic phone mount **700** can include a ferromagnetic base **792**, a ferromagnetic ring **733** positioned around the wireless charging opening **715**, and a ferromagnetic alignment element **759**. The ferromagnetic base **792** can be a continuous sheet of ferromagnetic material that occupies at least the locations **251** of the ferromagnetic base **792** that are directly opposite to where the permanent magnetic elements **250** of the personal storage container **105** will be located when magnetically coupled to the magnetic phone mount **700** using the ferromagnetic alignment element **759**.

[0051] FIG. **7C** is a schematic illustration of the magnetic phone mount **700** of FIG. **7A** coupled to a mobile device **767** and configured to magnetically couple to the modular personal storage system **100** including the personal storage containers **105** and covers **110** disclosed herein in some embodiments according to the inventive concept.

[0052] FIG. **7D** is a schematic illustration of the modular personal storage system **100** including a personal storage container **105** and cover **110** magnetically coupled to the magnetic phone mount **700** on the mobile device **767** in some embodiments according to the inventive concept.

[0053] As used herein any reference to “one embodiment” or “an embodiment” means that a particular element, feature, structure, or characteristic described in connection with the embodiment is included in at least one embodiment. The appearances of the phrase “in one embodiment” in

various places in the specification are not necessarily all referring to the same embodiment.

[0054] Some embodiments may be described using the expression “coupled” and “connected” along with their derivatives. For example, some embodiments may be described using the term “coupled” to indicate that two or more elements are in direct physical or electrical contact. The term “coupled,” however, may also mean that two or more elements are not in direct contact with each other, but yet still co-operate or interact with each other. The embodiments are not limited in this context.

[0055] As used herein, the terms “comprises,” “comprising,” “includes,” “including,” “has,” “having” or any other variation thereof, are intended to cover a non-exclusive inclusion. For example, a process, method, article, or apparatus that comprises a list of elements is not necessarily limited to only those elements but may include other elements not expressly listed or inherent to such process, method, article, or apparatus. Further, unless expressly stated to the contrary, “or” refers to an inclusive or and not to an exclusive or. For example, a condition A or B is satisfied by any one of the following: A is true (or present) and B is false (or not present), A is false (or not present) and B is true (or present), and both A and B are true (or present).

[0056] In addition, use of the “a” or “an” are employed to describe elements and components of the embodiments herein. This is done merely for convenience and to give a general sense of various embodiments. This description should be read to include one or at least one and the singular also includes the plural unless it is obvious that it is meant otherwise.

[0057] It will be appreciated that elements in the figures are illustrated for simplicity and clarity and have not necessarily been drawn to scale. For example, the dimensions and/or relative positioning of some of the elements in the figures may be exaggerated relative to other elements to help to improve understanding of various embodiments of the present inventive concept. Also, common but well-understood elements that are useful or necessary in a commercially feasible embodiment are often not depicted in order to facilitate a less obstructed view of these various embodiments. The same reference numbers may be used to describe like or similar parts. Further, while several examples have been disclosed herein, any features from any examples may be combined with or replaced by other features from other examples. Moreover, while several examples have been disclosed herein, changes may be made to the disclosed examples within departing from the scope of the claims.

[0058] Those skilled in the art will recognize that a wide variety of modifications, alterations, and combinations can be made with respect to the above described embodiments without departing from the scope of the inventive concept, and that such modifications, alterations, and combinations are to be viewed as being within the ambit of the inventive concept.

[0059] The terminology used herein is for the purpose of describing embodiments only and is not intended to be limiting to other embodiments. As used herein, the singular forms “a,” “an” and “the” are intended to include the plural forms as well, unless the context clearly indicates otherwise. It will be further understood that the terms “comprises,” “comprising,” “includes” and/or “including,” “have” and/or “having” when used herein, specify the presence of stated features, integers, steps, operations, elements, and/or components, but do not preclude the presence or addition of one or more other features, integers, steps, operations, elements, components, and/or groups thereof. Elements described as being “to” perform functions, acts and/or operations may be configured to or other structured to do so.

[0060] Unless otherwise defined, all terms (including technical and scientific terms) used herein have the same meaning as commonly understood by one of ordinary skill in the art to which various embodiments described herein belong. It will be further understood that terms used herein should be interpreted as having a meaning that is consistent with their meaning in the context of this specification and the relevant art and will not be interpreted in an idealized or overly formal sense unless expressly so defined herein.

[0061] Any terms generally associated with circles, such as “radius” or “radial” or “diameter” or

“circumference” or “circumferential” or any derivatives or similar types of terms are intended to be used to designate any corresponding structure in any type of geometry, not just circular structures. For example, “radial” as applied to another geometric structure should be understood to refer to a direction or distance between a location corresponding to a general geometric center of such structure to a perimeter of such structure; “diameter” as applied to another geometric structure should be understood to refer to a cross sectional width of such structure; and “circumference” as applied to another geometric structure should be understood to refer to a perimeter region. Nothing in this specification or drawings should be interpreted to limit these terms to only circles or circular structures.

[0062] Terms such as “substantially,” “about,” “approximately” or the like as used in referring to a relationship between two objects is intended to reflect not only an exact relationship but also variances in that relationship that may be due to various factors such as the effects of environmental conditions, common error tolerances, manufacturing variances, or the like. It should further be understood that although some values or other relationships may be expressed herein without a modifier, these values or other relationships may also be exact or may include a degree of variation due to various factors such as the effects of environmental conditions, common error tolerances, manufacturing variances, or the like.

[0063] In some embodiments, the term “about” generally refers to a range of numeric values that one of skill in the art would consider equivalent to the recited numeric value or having the same function or result. For example, “about” may refer to a range that is within $\pm 1\%$, $\pm 2\%$, $\pm 5\%$, $\pm 7\%$, $\pm 10\%$, $\pm 15\%$, or even $\pm 20\%$ of the indicated value, depending upon the numeric values that one of skill in the art would consider equivalent to the recited numeric value or having the same function or result. Furthermore, in some embodiments, a numeric value modified by the term “about” may also include a numeric value that is “exactly” the recited numeric value. In addition, any numeric value presented without modification will be appreciated to include numeric values “about” the recited numeric value, as well as include “exactly” the recited numeric value. Similarly, the term “substantially” means largely, but not wholly, the same form, manner or degree and the particular element will have a range of configurations as a person of ordinary skill in the art would consider as having the same function or result. When a particular element is expressed as an approximation by use of the term “substantially,” it will be understood that the particular element forms another embodiment.

[0064] Conditional language, such as, among others, “can,” “could,” “might,” or “may,” unless specifically stated otherwise, or otherwise understood within the context as used, is generally intended to convey that certain embodiments include, while other embodiments do not include, certain features, elements, and/or steps. Thus, such conditional language is not generally intended to imply that features, elements and/or steps are in any way required for one or more embodiments or that one or more embodiments necessarily include logic for deciding, with or without user input or prompting, whether these features, elements and/or steps are included or are to be performed in any particular embodiment.

[0065] Unless the context clearly requires otherwise, throughout the description and the claims, the words “include,” “can include,” and the like are to be construed in an inclusive sense, as opposed to an exclusive or exhaustive sense; that is to say, in the sense of “including, but not limited to.” As used herein, the terms “connected,” “coupled,” or any variant thereof means any connection or coupling, either direct or indirect, between two or more elements; the coupling or connection between the elements can be physical, logical, or a combination thereof. Additionally, the words “herein,” “above,” “below,” and words of similar import, when used in this application, refer to this application as a whole and not to any particular portions of this application. Where the context permits, words in the above Detailed Description using the singular or plural number may also include the plural or singular number respectively. The word “or” in reference to a list of two or more items, covers all of the following interpretations of the word: any one of the items in the list,

all of the items in the list, and any combination of the items in the list. Likewise the term “and/or” in reference to a list of two or more items, covers all of the following interpretations of the word: any one of the items in the list, all of the items in the list, and any combination of the items in the list.

[0066] Disjunctive language such as the phrase “at least one of X, Y, or Z,” unless specifically stated otherwise, is otherwise understood with the context as used in general to present that an item, term, etc., may be either X, Y, or Z, or any combination thereof (non-limiting examples: X, Y, and/or Z). Thus, such disjunctive language is not generally intended to, and should not, imply that certain embodiments require at least one of X, at least one of Y, or at least one of Z to each be present.

[0067] Unless otherwise explicitly stated, articles such as “a” or “an” should generally be interpreted to include one or more described items. Accordingly, phrases such as “a device configured to” are intended to include one or more recited devices. Such one or more recited devices can also be collectively configured to carry out the stated recitations. For example, “a processor configured to carry out recitations A, B and C” can include a first processor configured to carry out recitation A working in conjunction with a second processor configured to carry out recitations B and C.

[0068] While the Detailed Description f has shown, described, and pointed out novel features as applied to various embodiments, it can be understood that various omissions, substitutions, and changes in the form and details of the devices or algorithms illustrated can be made without departing from the spirit of the disclosure. As can be recognized, certain embodiments described elsewhere herein can be embodied within a form that does not provide all of the features and benefits set forth herein, as some features can be used or practiced separately from others. The scope of certain embodiments disclosed herein is indicated by the appended claims rather than by the foregoing description. All changes which come within the meaning and range of equivalency of the claims are to be embraced within their scope.

[0069] These and other changes can be made to the invention in light of the Detailed Description. While the above description describes certain examples of the invention, and describes the best mode contemplated, no matter how detailed the above appears in text, the invention can be practiced in many ways. Details of the system may vary considerably in its specific implementation, while still being encompassed by the invention disclosed herein. As noted above, particular terminology used when describing certain features or aspects of the invention should not be taken to imply that the terminology is being redefined herein to be restricted to any specific characteristics, features, or aspects of the invention with which that terminology is associated. In general, the terms used in the following claims should not be construed to limit the invention to the specific examples disclosed in the specification, unless the above Detailed Description section explicitly defines such terms. Accordingly, the actual scope of the invention encompasses not only the disclosed examples, but also all equivalent ways of practicing or implementing the invention under the claims.

Claims

1. A modular personal storage system comprising: a personal storage container including: a first outer wall including a paramagnetic metal or a diamagnetic metal; a second outer wall including the paramagnetic metal or the diamagnetic metal, the second outer wall positioned opposite to and spaced apart from the first outer wall to define a container volume between the first outer wall and the second outer wall; and at least two permanent magnetic elements spaced apart from one another and at least partially located within the container volume.
2. The modular personal storage system of claim 1 wherein the paramagnetic metal or the diamagnetic metal includes aluminum.

3. The modular personal storage system of claim 1 wherein the paramagnetic metal or the diamagnetic metal allows a magnetic field to pass through the second outer wall into the container volume without being substantially reduced.
4. The modular personal storage system of claim 1 wherein a total distance from an outer surface of the first outer wall to an outer surface of the second outer wall is in a range between about 3.8 mm to about 5.5 mm.
5. The modular personal storage system of claim 1 wherein the at least two permanent magnetic elements include first and second opposite poles, positioned in the container volume with the first poles facing the first outer wall and the second poles facing the second outer wall.
6. The modular personal storage system of claim 5 wherein surfaces of the first and second poles are located about equal distances from the first and second outer walls, respectively.
7. The modular personal storage system of claim 1 wherein the first outer wall includes a first alignment opening that exposes the container volume.
8. The modular personal storage system of claim 7 wherein the second outer wall includes a second alignment opening that exposes the container volume, the second alignment opening being aligned to the first alignment opening in two dimensions.
9. The modular personal storage system of claim 1 further comprising: an outer side wall that extends from an edge of the first outer wall to an edge of the second outer wall and to define an access opening to inside the container volume; an interior side wall that extends from the first outer wall to the second outer wall inside the container volume; and first and second deformable bumpers protruding from opposing portions of the interior side wall toward one another and configured to securely store a plurality of payment cards inside the container volume.
10. The modular personal storage system of claim 9 further comprising: a lever extending from inside the container volume through the outer side wall to outside the container volume; a hinge, inside the container volume and coupled to the lever; and a finger, inside the container volume and coupled to the hinge.
11. The modular personal storage system of claim 10 wherein the finger includes a staircase shaped edge configured to contact edges of the plurality of payment cards when the lever is moved about the hinge to rotate the finger toward the access opening so that the staircase shaped edge pushes the payment cards between the first and second deformable bumpers through the access opening in the container volume.
12. The modular personal storage system of claim 11 wherein the staircase shaped edge includes a number of steps each configured to separately contact a respective card in the container volume.
13. The modular personal storage system of claim 1 wherein the personal storage container is configured for inclusion in a stack of a plurality of personal storage containers having contacting ones of the personal storage containers in the stack magnetically coupled together via contacting ones of the first and second outer walls of first and second personal storage containers.
14. The modular personal storage system of claim 1 wherein the personal storage container comprises a first personal storage container, the modular personal storage system further comprising: a second personal storage container, configured to magnetically coupled to the first personal storage container, the second personal storage container including: a first outer wall that is spaced apart from a second outer wall to define a second container volume between the first outer wall and the second outer wall of the second personal storage container; and at least two permanent magnetic elements at least partially located within the second container volume adjacent to the first outer wall of the second personal storage container, wherein first poles of the at least two permanent magnet elements in the second personal storage container face the second outer wall of the first personal storage container and are configured to generate a magnetic field that extends through the first outer wall of the second personal storage container through the second outer wall of the first personal storage container to magnetically couple to the at least two permanent magnetic elements of the first personal storage container.

15. The modular personal storage system of claim 1 further comprising: a first pair of separator bumpers each protruding from a surface of the first outer wall by a first amount of about 0.15 mm; and a second pair of separator bumpers each protruding from a surface of the second outer wall by a second amount of about 0.15 mm.

16. The modular personal storage system of claim 1 wherein the paramagnetic metal or the diamagnetic metal includes copper, titanium, or magnesium.

17. The modular personal storage system of claim 8 wherein the first opening and the second opening define first and second stadium shapes, respectively, each having a strait portion length of about 16.5 mm and a curved portion radius of about 0.9 mm.

18. The modular personal storage system of claim 1 wherein the at least two permanent magnetic elements comprise at least six permanent magnetic elements spaced apart along three contiguous outer side walls of an outer perimeter of the container volume.

19. A modular personal storage kit comprising: a personal storage container including: a first outer wall including a paramagnetic metal or a diamagnetic metal; a second outer wall including the paramagnetic metal or the diamagnetic metal, the second outer wall positioned opposite to and spaced apart from the first outer wall to define a container volume between the first outer wall and the second outer wall; and at least two permanent magnetic elements spaced apart from one another and at least partially located within the container volume; and a removeable cover for the personal storage container, configured to magnetically couple to the personal storage container, the removeable cover including: a cover inner surface configured to face the second outer wall of the personal storage container when the removeable cover is removably coupled to the personal storage container; a cover outer surface, opposite the cover inner surface, the cover outer surface configured to be exposed when the removeable cover is removably coupled to the personal storage container; and a ferromagnetic pattern located between the cover inner surface and the cover outer surface, the ferromagnetic pattern configured to magnetically couple the removeable cover to the personal storage container via the at least two permanent magnetic elements.

20. The modular personal storage kit of claim 19 further comprising: a stadium shaped opening in the second outer wall of the personal storage container that exposes the container volume; and a stadium shaped alignment protrusion protruding from a location on cover inner surface corresponding to that of the stadium shaped opening of the personal storage container configured to securely fit into the stadium shaped opening when the removeable cover is removably coupled to the personal storage container.

21. The modular personal storage kit of claim 20 wherein dimensions of the stadium shaped opening are about 0.3 mm greater than dimensions of the stadium shaped alignment protrusion.

22. The modular personal storage kit of claim 19 wherein the ferromagnetic pattern between the cover inner surface and the cover outer surface is positioned to overlap the least two permanent magnetic elements.

23. The modular personal storage kit of claim 19 wherein the ferromagnetic pattern forms first and second open facing U-shaped patterns at opposite ends of the removeable cover.

24. The modular personal storage kit of claim 23 wherein the removeable cover further includes: first and second notches in opposing sides of the removeable cover; and an elastic member extending across the cover outer surface to the cover inner surface through the first and second notches.

25. The modular personal storage kit of claim 19 wherein the removeable cover further comprises: a recessed tray in the cover outer surface the recessed tray having a payment card shape and having a bottom surface; a tray edge extending along a perimeter of the recessed tray above the bottom surface; and a tray lip extending from an upper surface of the tray edge opposite the bottom surface of the recessed tray.

26. The modular personal storage kit of claim 25 further including: first and second elongated tray skids protruding from the bottom surface of the recessed tray and extending along a major

dimension of the recessed tray spaced apart.

27. A handheld personal storage container for releasably storing a set of payment cards and for magnetically removably coupling to a portable mobile device and other similar personal storage containers, the container including: a first outer wall including a paramagnetic metal or a diamagnetic metal; a second outer wall including the paramagnetic metal or the diamagnetic metal, the second outer wall positioned opposite to and spaced apart from the first outer wall; an outer side wall coupled to an edge of the first outer wall and an edge of the second outer wall and extending around a perimeter of the first outer wall and the second outer wall to form an opening in the outer side wall; a container volume defined to be between the first outer wall and the second outer wall within the perimeter; at least six permanent magnetic elements located separated from one another along the perimeter; first and second deformable bumpers located on opposing sides of the container volume; a lever extending from inside the container volume through the outer side wall to outside the container volume; a hinge, inside the container volume and coupled to the lever; a finger, inside the container volume and coupled to the hinge; and a staircase shaped edge on the finger configured to contact edges set of payment cards when the lever is moved about the hinge to rotate the finger toward the opening so that the staircase shaped edge pushes the set of payment cards between the first and second deformable bumpers through the opening in the container volume.
